

Ammar Eltigani

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Education

University of Michigan , Department of Mathematics, Ann Arbor, MI	2024–present
Ph.D. student in Mathematics	2026–present
M.S. in Mathematics	2026
Williams College , Williamstown, MA	2019–2024
B.A. in Mathematics and Computer Science	Jan. 2024
University of Oxford , Oxford, UK	2021–2022
Williams–Exeter Programme at Oxford Fellow	

Research Interests

Random matrix theory and high-dimensional probability; additive combinatorics and convex geometry; commutative algebra.

Papers, Manuscripts, and Expository Writing

- [1] *The Khovanskii polynomial in $\mathbb{Z}^d$* . Undergraduate thesis, Williams College, 2024. Advisor: Leo Goldmakher.
- [2] *Completions of quasi-excellent domains*. SMALL REU project, Williams College, 2023. Advisor: Susan Loepp.
- [3] *Random Matrix Theory for High-Dimensional Machine Learning*. Expository article/talk notes, 2026.

Research Experience

Undergraduate Thesis, Williams College 2023–2024
The Khovanskii polynomial in $\mathbb{Z}^d$. Advisor: Leo Goldmakher.

- Studied Khovanskii polynomials associated to finite subsets of $\mathbb{Z}^d$, drawing on additive combinatorics, Ehrhart theory, lattice residues, and convex geometry.
- Derived an explicit binomial formula for the Khovanskii polynomial of a finite set $A \subset \mathbb{Z}^d$.
- Established bounds for phase transitions of the Khovanskii polynomial for families of sets.

SMALL Research Experience for Undergraduates, Williams College Summer 2023
Advisor: Susan Loepp.

- Investigated completions of local rings and quasi-excellent domains.
- Characterized completions of quasi-excellent domains and studied related formal fiber conditions.
- Presented the work at the Young Mathematician’s Conference and JMM 2024.

Research Software Engineering Internship, Google Quantum AI Summer 2022

- Added a feature to Google’s open-source quantum circuit compiler Cirq ([link](#)).
- Designed ([link](#)) and implemented ([link](#)) a qubit routing algorithm based on [arxiv:1902.08091](#).
- Improved performance and runtime of existing implementation by 100x, outperforming IBM and `t|ket>`.

Selected Talks

<i>Random Matrix Theory for High-Dimensional Machine Learning.</i> Marjorie Lee Browne Symposium, University of Michigan	2026
<i>Stability of the Size of Sumsets.</i> Combinatorics Student Seminar, University of Michigan	2025
<i>Controlling Formal Fibers of Countably Many Principal Prime Ideals.</i> JMM	2024
<i>Completions of Quasi-Excellent Domains.</i> Joint Mathematics Meetings	2024
<i>Completions of Quasi-Excellent Domains.</i> Young Mathematician's Conference	2023

Conferences, Workshops, and Schools

University of Michigan Summer School in Random Matrix Theory	2026
BRIDGES: Graphical Designs, Algebraic Statistics, and Geometric Group Theory	2024
Joint Mathematics Meetings	2024
Young Mathematician's Conference	2023

Teaching

Graduate Student Instructor, University of Michigan 2024–present
Math 115/116: Calculus I/II. Primary instructor for introductory calculus sections; duties include lecturing, preparing quizzes, grading exams, and holding office hours.

Graduate Student Assistant, MaCSS Summer Institute, University of Michigan Summer 2025
Taught a probability crash course, mentored undergraduate scholars, and led weekly group study sessions.

Teaching Assistant, Williams College 2020–2023
Led help sessions and graded homework for courses including Applied Real Analysis and Abstract Algebra.

Honors and Fellowships

Herschel–Smith Fellowship	2024
Williams–Exeter Programme at Oxford Fellow	2021–2022

Service and Mentoring

Undergraduate mentor, MaCSS Summer Institute, University of Michigan	2025
Tutor, Underrepresented Identities in Computer Science, Williams College	2021–2023